

Total No. of Questions : 8]

SEAT No. :

PD-4047

[Total No. of Pages : 3

[6402]-6

S.E. (Civil Engg.)

GEOTECHNICAL ENGINEERING

(2019 Pattern) (Semester - IV) (201008) (Theory)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary and mention it clearly.
- 5) Use of non-programmable calculator is allowed.

- Q1) a)** Describe the effect of compaction on properties of soil. [5]
- b) Explain how compaction control is achieved in the field using a proctor needle. [6]
- c) What is pressure bulb? Explain its significance and draw a neat sketch of pressure bulb for concentrated point load. [6]

OR

- Q2) a)** Explain Boussinesq's theory with neat sketch. State its Assumptions and formula for stress in soil by point load, with description of each term. [6]
- b) Differentiate between Light Compaction & Heavy Compaction test. [5]
- c) A load of 1000 kN acts as a point load at the surface of a soil mass. Estimate the stress at a point 3m below and 4m away from the point of action of the load by Boussinesq's formula. Compare the value with the result from Westergaards's theory. [6]

- Q3) a)** Explain how shear tests are conducted with different drainage conditions? [5]
- b) Explain vane shear test procedure with a neat sketch and formula. [6]
- c) Define total and effective stress. Determine the shear strength in terms of effective stress on a plane within a saturated soil mass at a point where the total normal stress is 200 KN/m² and the pore water pressure is 80 KN/m². The effective stress shear strength parameters for the soil are $c' = 16$ KN/m² and $\phi = 39^\circ$. [6]

P.T.O.

OR

- Q4)** a) A sample of dry cohesionless soil was tested in a triaxial machine. If the angle of internal shearing resistance was 36° & the confining pressure 100kN/m^2 , determine the deviator stress at which the sample failed. [6]
- b) Describe the procedure for Direct Shear Test. [5]
- c) Explain briefly the procedure of conducting Unconfined Compression Test on Clayey soil sample. Draw Mohr's circle for the test. [6]
- Q5)** a) Explain Rebhann's graphical method for determination of earth pressure on retaining wall. [6]
- b) Derive the expression for active state of pressure at any point for dry cohesionless backfill with uniform surcharge. [6]
- c) A smooth vertical wall retains a level surface with $\gamma = 18\text{kN/m}^3$, $\Phi = 30^\circ$, to a depth of 8 m. Draw the lateral pressure diagram and compute the total active pressure in dry condition and when water table rises to the GL. Assume $\gamma_{\text{sat}} = 22\text{kN/m}^3$. [6]

OR

- Q6)** a) Determine the relation for lateral earth pressure in active state for submerged cohesionless backfill. [6]
- b) Discuss Culmann's graphical method for the determination of active earth pressure. [6]
- c) In a cohesionless soil deposit having unit weight of 15 kN/m^3 and angle of internal friction Φ of 30° . Determine the active and passive lateral pressure intensities at depth of 10 m. [6]
- Q7)** a) Discuss "Taylor's Stability Number" for stability analysis of finite slope. [6]
- b) Enlist factors of safety used in stability analysis of slopes. Calculate the factor of safety w.r.t. cohesion of clay slope laid at 1 in 2 to a height of 10 m, if the angle of internal friction $\Phi = 10^\circ$ and $c = 25\text{kN/m}^2$ and $\gamma = 19\text{kN/m}^3$. What will be the critical height of the slope in this soil. Assume $S_n = 0.064$ for $\Phi = 10^\circ$. [6]
- c) Discuss the causes and remedial measures of Landslides. [6]

OR

- Q8)** a) Derive the expression for factor of safety for dry infinite slope in sandy soil. [6]
- b) Explain with neat sketch different modes of slope failure. [6]
- c) Discuss “Swedish Slip Circle Method” for stability analysis of finite slope. [6]
